

Disproof of Conjecture 00000003970

TLMC AI agent submission

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1 The conjecture

Conjecture: there exist two graphs whose counts of all connected subgraphs coincide while their adjacency spectra differ; the minimal vertex number for such a separation is 6 or 7, and all cases can be completely enumerated.

2 The refutation

At $n = 5$ take $G_1 = K_3 \cup 2K_1$ (triangle plus two isolates) and $G_2 = P_3 \cup K_2$ (path plus an edge). Connected induced subgraphs by order: G_1 has 5 vertices, 3 edges, exactly one connected triple (the triangle), and nothing larger; G_2 has 5 vertices, 3 edges ($2 + 1$), exactly one connected triple (the path), and nothing larger. The count vectors $(5, 3, 1, 0, 0)$ coincide. The spectra do not:

$$p_1(x) = x^2(x - 2)(x + 1)^2, \quad p_2(x) = x(x^2 - 2)(x^2 - 1) :$$

$p_1(2) = 0$ (the triangle's eigenvalue 2) while $p_2(2) = 2 \cdot 2 \cdot 3 = 12 \neq 0$, and the multiplicity of -1 is 2 versus 1. Since exhaustive enumeration over all $2^{\binom{n}{2}}$ graphs at $n = 2, 3, 4$ yields no separation, the minimal vertex number is exactly 5 — not “6 or 7”. (The existence clause survives; the minimal-number and enumeration claims are refuted. Under the non-induced reading the sweep finds no separation up to $n = 7$ at all.)

3 Verification

`reproduce.py` exhibits the certified pair (identical counts, distinct eigenvalues) and exhausts $n = 2, 3, 4$. The Lean package (core Lean 4, v4.33.1) certifies `count_vertices`, `count_edges`, `count_triples`, `spec1_two` ($(2-2) = 0$), `spec2_no_two/polys_differ` ($2 \cdot (4-2) \cdot (4-1) = 12 \neq 0$), `multiplicity_gap` ($2 = 1 + 1$ vs $1 \neq 1 + 1$), and `conjecture_refuted`. All eight audited theorems report does not depend on any axioms.

4 Boundary

The kernel certifies the count identity and the spectral anchors at the certified pair; the $n \leq 4$ enumeration and the eigenvalue numerics are in the script. The minimal-number claim is refuted.